

# KIRTI GONDALIYA

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## PROFESSIONAL SUMMARY

Experienced Data Engineer with 4 years of experience in designing and optimizing large-scale data pipelines, cloud infrastructure, and real-time analytics. Proficient in Hadoop, Apache Kafka, Spark, AWS, and database management, handling terabytes of healthcare and financial data. Skilled in ETL workflows, data governance, and compliance (HIPAA, GDPR). Adept at improving query performance, implementing machine learning pipelines, and enhancing system reliability. Passionate about leveraging data engineering to drive operational efficiency and actionable insights.

## EDUCATION

**Master of Science in Computer Science | GPA 3.8**

**January 2023 – December 2024**

Lamar University, Beaumont, Texas

**Bachelors in Electronics and Communication Engineering | CGPA 7.94**

**August 2017 – June 2021**

Shree Swami Atmanand Institute of Technology, Surat, Gujarat, India

## PROFESSIONAL EXPERIENCE

**Data Engineer | Humana, Remote, USA**

**Jan 2024 – Present**

- Optimized Hadoop clusters managing 10TB+ healthcare data, ensuring high availability and fault tolerance, reducing downtime by 30% and enhancing data access for clinical research teams.
- Scaled Cassandra databases with millions of patient records, implementing indexing and replication strategies, improving query performance by 40% for real-time patient data retrieval.
- Deployed AWS-based healthcare infrastructure with EC2 and 50TB S3 storage, enforcing encryption and access controls, streamlining security audits and reducing compliance risks.
- Built real-time data pipelines with Kafka and Spark Streaming, processing IoT data points daily, collaborating with data science and IoT teams to enhance predictive analytics.
- Developed ETL workflows for EMR and wearables, integrating HL7/FHIR standards, improving data interoperability and reducing integration failures by 50% across hospital networks.
- Integrated BI tools with Hadoop, creating KPI-driven dashboards for hospital operations, working with healthcare admins and clinical analysts to enhance patient care insights.
- Built Elasticsearch-based logging and monitoring, automating Logstash ingestion, reducing incident resolution time by 25% with real-time Kibana dashboards.

**Data Engineer | Sigma InfoSolutions, Bangalore, India**

**May 2020 – Dec 2022**

- Built a near-real-time data pipeline using Confluent Kafka, Debezium, AWS MSK, and Snowflake to ingest 100M rows daily, optimizing fraud detection and reducing processing lag by 40%.
- Utilized TextractOCR, Spark on EMR, S3, Apache Airflow, and Dagster with CI/CD via CodePipeline and CodeBuild, improving deployment efficiency and increasing data throughput.
- Revamped Medicare and Medicaid data pipelines using Spark on EMR and NLP on SageMaker, creating a denormalized DynamoDB store that enhanced query response times by 60%.
- Partnered with cross-functional teams to scale data ingestion from 10M to 250M rows per day, implementing robust and scalable processing solutions that reduced failure rates.
- Optimized SQL queries for Presto, AWS Athena, and Snowflake, reducing execution times by 30%, leading to faster insights and enabling real-time decision-making for analytics teams.
- Implemented AWS Glue Data Quality, achieving 95% accuracy in data validation and cleansing processes, improving overall data quality and governance while reducing compliance risks.
- Collaborated with data engineering, DevOps, and product teams to streamline ETL workflows and enhance system reliability.

## TECHNICAL SKILLS

**Programming Languages**

Python, SQL, R, SCALA, MySQL, MS SQL Server

**Big Data & Data Engineering Tools**

Apache Airflow, Debezium, Cassandra, Kafka, Delta Lake

**Cloud Platforms & Services**

AWS (EC2, S3, Lambda, CloudFormation, EMR), GCP, Pulumi, Google Bigtable

**Data Warehousing & ETL Tools**

Snowflake, Amazon DynamoDB, AWS Redshift, Hadoop, AWS Glue, Apache NiFi, Apache Spark, Apache Hudi, Iceberg

**Data Analysis & Visualization**

Tableau, Power BI, Kibana (Elasticsearch), Matplotlib, Seaborn, Plotly, Grafana, Pandas & NumPy, ML Frameworks (TensorFlow, PyTorch)

**Version Control & Development Tools**

Git, GitHub, AWS CodeCommit, Jenkins, Docker, Postman, Kubernetes, Terraform, PySpark / Jupyter Notebook, Apache Airflow / Dagster, AWS CodePipeline & CodeBuild

**Operating Systems**

Windows, macOS

## PROJECT

**Project Name:** Patient Data Integration and Real-Time Analytics

**Technologies:** Apache Kafka, Spark Streaming, AWS (EC2, S3), Cassandra

- Developed a real-time data pipeline to ingest and process IoT patient data streams, ensuring high availability and fault tolerance.
- Integrated data from wearable devices into existing Cassandra databases, enabling real-time patient health tracking and reducing data latency by 40%.
- Built custom Spark jobs to enrich incoming data and apply transformations, improving data quality for downstream analytics.

## CERTIFICATIONS

- AWS Data Engineer
- AWS Certified Solution Architect